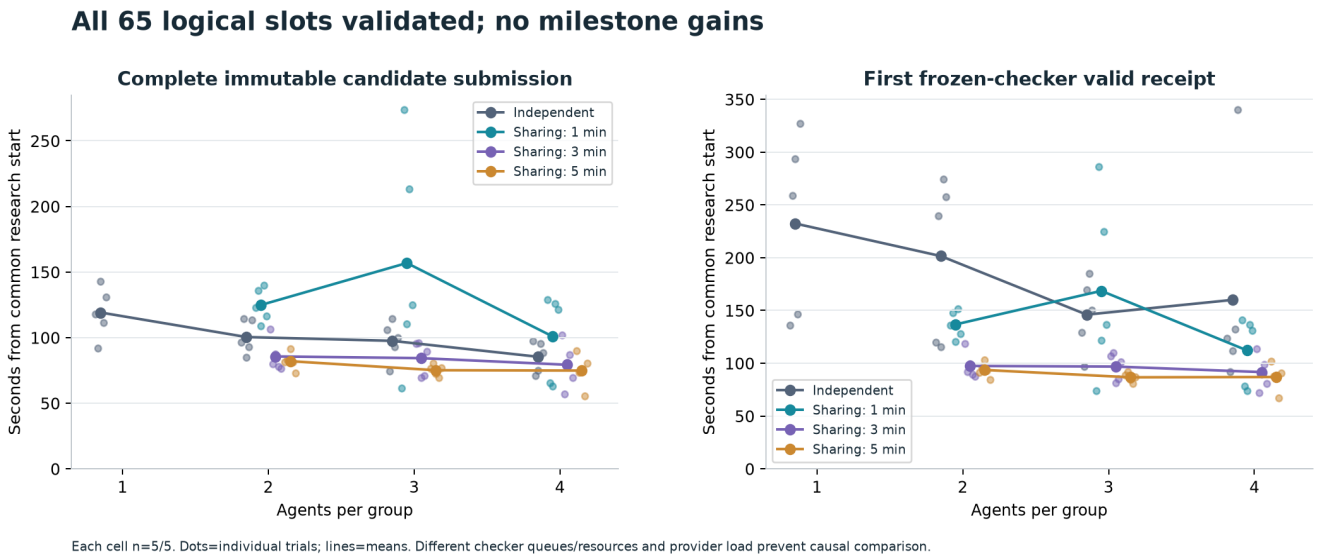
Valid constructions, no new gains

20 independent trials and 45 fast shared-context trials

All 65 logical slots passed the frozen checker and independent revalidation. Every primary accepted word pair was byte-identical to the supplied baseline. No trial reached a requested improvement milestone. The stopping rule explicitly allowed baseline eligibility.

INDEPENDENT	FAST SHARING	MILESTONE GAINS
20/20 valid	45/45 valid	0/65 trials



Each cell n=5/5. Dots=individual trials; lines=means. Different checker queues/resources and provider load prevent causal comparison.

What the experiment establishes

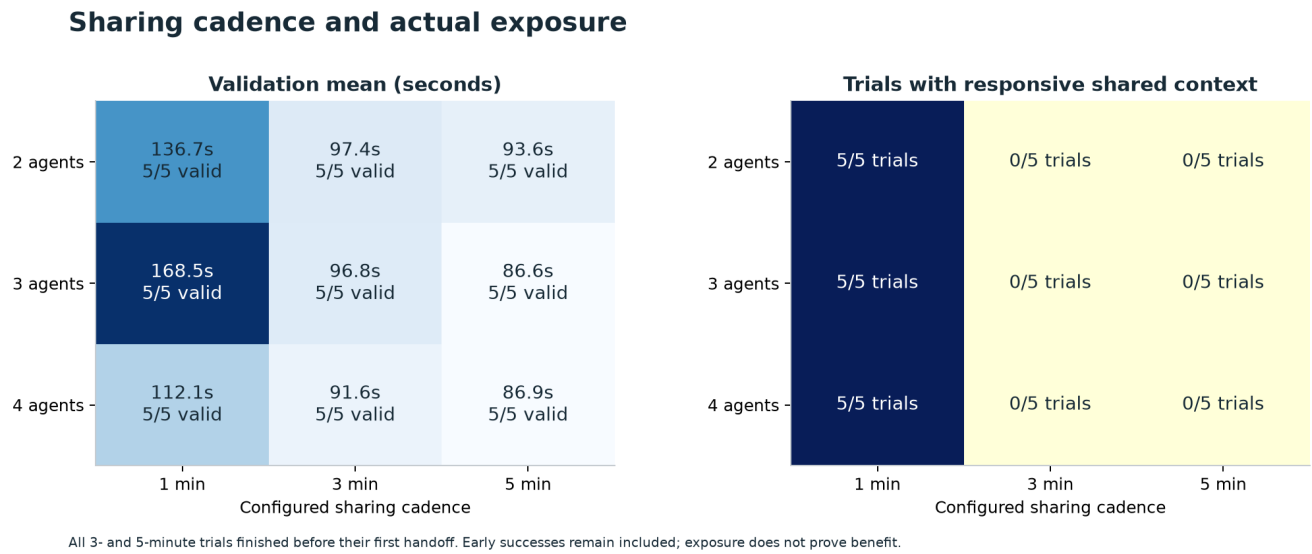
The fast transport worked: checkpoints were written while agents worked, captured after draining, literally combined without a synthesis model, and injected into fresh same-group sessions. **15/45 sharing trials received responsive shared context.** Thirty finished before any handoff.

It does **not** establish that sharing, prompt diversity or more agents caused better research. The outcome was baseline reproduction, proposed approaches overlapped, and checker admission waits differed sharply between arms. Timing charts are descriptive, with all five trials retained in every cell.

One capacity-failed attempt was retained separately and replaced once with fresh workers under user authorization. All 65 logical slots are successful; the 46th sharing attempt is infrastructure history, excluded from performance means. No successful trial was rerun.

Cadence and actual exposure

Five planned and successful trials in every count-by-cadence cell



All 3- and 5-minute trials finished before their first handoff. Early successes remain included; exposure does not prove benefit.

Agents	Cadence	Valid	Submit mean	Validate mean	Responsive context
2	1 min	5/5	124.9 s	136.7 s	5/5
2	3 min	5/5	85.6 s	97.4 s	0/5
2	5 min	5/5	82.0 s	93.6 s	0/5
3	1 min	5/5	156.8 s	168.5 s	5/5
3	3 min	5/5	84.3 s	96.8 s	0/5
3	5 min	5/5	75.1 s	86.6 s	0/5
4	1 min	5/5	100.8 s	112.1 s	5/5
4	3 min	5/5	79.3 s	91.6 s	0/5
4	5 min	5/5	74.9 s	86.9 s	0/5

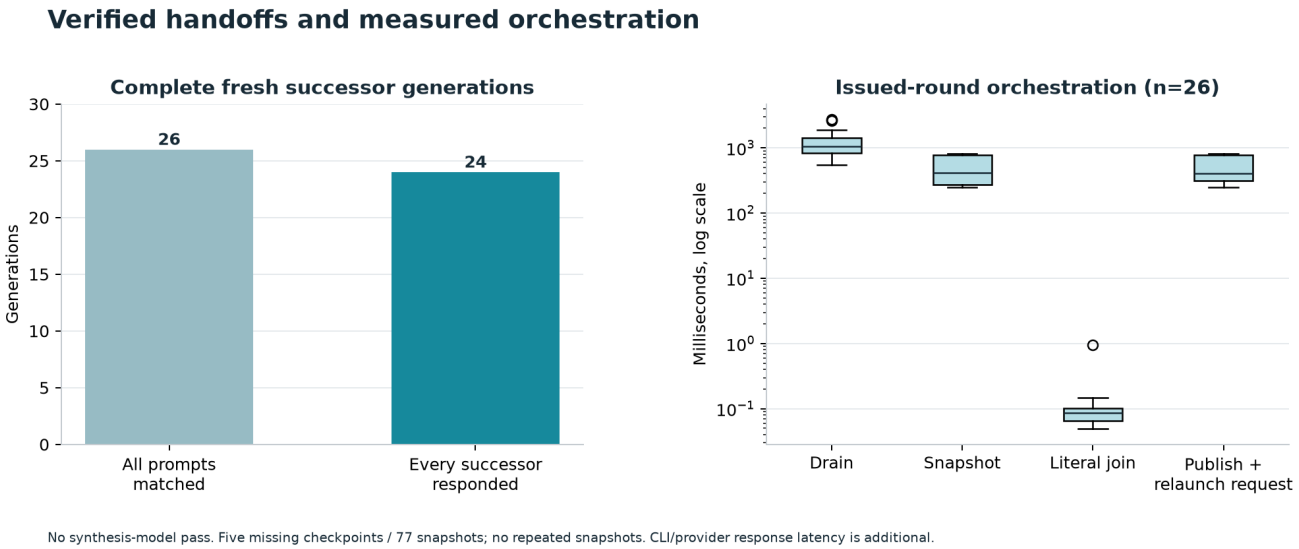
All 30 trials configured for 3- or 5-minute sharing finished before the first handoff. Their labels describe a configured sharing policy, not actual context exposure. All 15 one-minute trials received at least one fully response-corroborated generation.

The lowest observed cell mean was the four-agent / five-minute policy: 74.9 s to immutable submission and 86.9 s to validation. It performed zero handoffs. That observation cannot support a sharing benefit. The one-minute arms had more relaunch overhead and also include trials that took longer to submit.

Complete numerical summaries for both submission and validation - mean, median, sample SD and range - appear on page 5 and in final-cell-statistics.csv. No failed-attempt elapsed time enters these means.

Literal handoffs and response

Separate a prompt receipt from a model that actually responds



No synthesis-model pass. Five missing checkpoints / 77 snapshots; no repeated snapshots. CLI/provider response latency is additional.

26 generations had all fresh prompts and context hashes matched; 24 also had a response from every successor. In two boundary generations, one or more successors were stopped without a response. These remain recorded prompt receipts, not evidence of completed productive research.

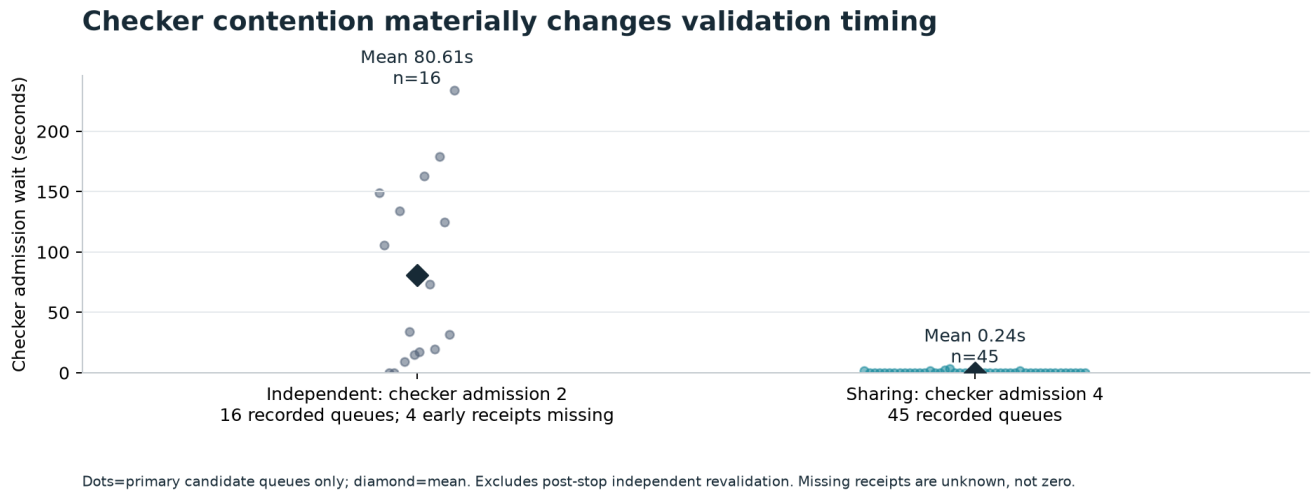
Phase, issued rounds n=26	Mean (ms)	Median (ms)	Range (ms)
Drain current work	1239.164	1046.600	549.420-2717.288
Capture checkpoint files	514.798	409.453	249.092-808.751
Literal file combination	0.120	0.086	0.049-0.942
Publish + relaunch request	529.652	404.892	249.248-816.014
Total to relaunch request	2283.733	2192.030	1047.864-4296.740

The file join averaged **0.120 ms**. Orchestration averaged **2.284 s** before actual CLI/provider response latency. For 73 responsive successor sessions, recorded prompt preparation to first model response averaged **5.62 s** (median 5.33; range 3.87-9.15). These session-level and round-level intervals can overlap; they are not simply additive and do not isolate pure provider startup.

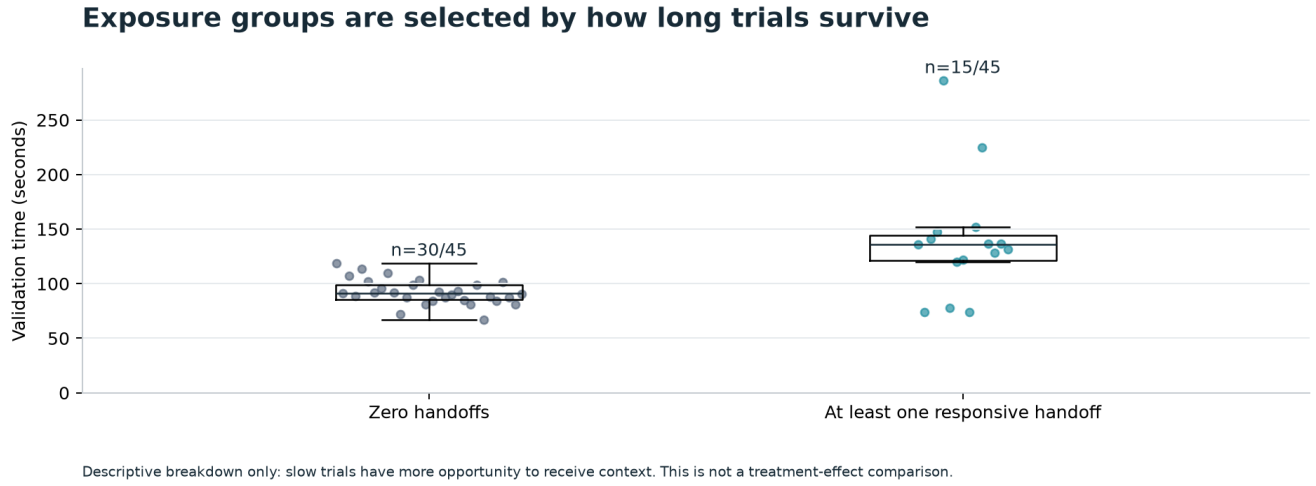
Checkpoint snapshots:5 missing out of 77 requested source snapshots;0 repeated according to recorded snapshot hashes. All 26 literal combinations were reproduced byte-for-byte. Model-written checkpoint statements were treated as untrusted research data.

Queues and selection effects

Resources and survival time confound the apparent timing advantage



Independent primary validation queues averaged **80.61 s across 16 recorded queues**; sharing averaged **0.24 s across 45**. Four early independent trials have no admission receipt; their queue waits are unknown. They are not filled with zeros. Post-stop independent revalidation is excluded from this queue comparison.



Longer-running trials have more opportunities to reach a sharing boundary. Comparing exposed and unexposed groups is therefore selected by survival time and cannot estimate a sharing treatment effect. No forced wait or higher novelty threshold was imposed to create more handoffs.

Outside research clocks: creation-to-ready elapsed time for 139 sharing workers averaged 16.38 s (median 9.64; range 7.42-394.39). This includes setup/control delays, not just provisioning CPU. Every group also had a 300 s cooldown. Parallel cooldowns overlap: their sums are not experiment wall time. Checker queues and handoff overhead remain inside research clocks.

Complete timing statistics

Seconds from common group start; every row is 5/5 valid

Complete immutable candidate submission

Arm / agents	Cadence	Mean	Median	Sample SD	Range
Independent /1	-	119.1	117.9	19.3	92.2-143.0
Independent /2	-	100.4	96.3	13.0	85.2-114.4
Independent /3	-	97.5	100.0	15.0	74.7-114.2
Independent /4	-	85.4	88.7	12.1	70.7-97.2
Sharing /2	1 min	124.9	122.9	13.1	108.9-140.0
Sharing /2	3 min	85.6	80.0	12.5	76.3-106.6
Sharing /2	5 min	82.0	81.5	6.4	73.1-91.2
Sharing /3	1 min	156.8	125.0	85.3	61.5-273.8
Sharing /3	3 min	84.3	89.6	13.1	69.4-96.1
Sharing /3	5 min	75.1	76.5	4.3	69.3-80.6
Sharing /4	1 min	100.8	121.2	33.6	63.2-128.8
Sharing /4	3 min	79.3	81.0	17.0	57.1-102.0
Sharing /4	5 min	74.9	74.7	12.7	55.4-90.2

First frozen-checker valid receipt

Arm / agents	Cadence	Mean	Median	Sample SD	Range
Independent /1	-	232.4	258.7	86.8	136.0-327.2
Independent /2	-	201.5	239.5	77.4	115.7-274.5
Independent /3	-	146.0	150.0	34.7	96.6-185.2
Independent /4	-	160.0	123.8	101.9	91.9-340.2
Sharing /2	1 min	136.7	135.8	13.1	120.3-151.8
Sharing /2	3 min	97.4	91.8	12.6	87.9-118.6
Sharing /2	5 min	93.6	93.4	6.8	84.5-103.2
Sharing /3	1 min	168.5	136.5	85.4	73.7-285.9
Sharing /3	3 min	96.8	101.4	13.0	81.2-109.6
Sharing /3	5 min	86.6	87.2	4.3	81.0-92.1
Sharing /4	1 min	112.1	131.3	33.2	73.9-140.9
Sharing /4	3 min	91.6	92.6	16.0	72.2-113.5
Sharing /4	5 min	86.9	87.4	12.5	67.2-101.9

n=5 in every summary; sample SD uses n-1. Submission means complete immutable snapshot receipt, not the agent-written manifest timestamp. Validation is first checker acceptance; the later independent replay is a separate validity audit. All 65 slots succeeded, so 3600 s cap-penalized means equal the displayed means. No cap penalty was applied to the separate capacity failure.

Full audit and model usage

Validity passed; prompt adherence and scientific claim quality have limits

Evidence	Independent	Sharing
Logical validated slots	20/20	45/45
Research worker instances	50	139
Actual unique provider sessions	50	216
Manually read tool inputs	517/517	1,006/1,006
Terminal plan files, each 3 proposals	49/50	136/139
Initial plans written after peer context	Not applicable	5 workers
Missing usage sessions	0/50	19/216 unknown

Actual initial user/developer contexts, fresh IDs, model/high effort, workspace-write policy, disabled network/memory, baseline lineage and source hashes were checked against fully reviewed references. All 139 sharing terminal exports matched their immutable receipts; all 216 IDs matched result records. No prohibited cross-trial/history/solution reads, extra agent model calls, web or MCP use were observed in the full input review. Built-in skill listings and generic collaboration instructions were advertised but unused.

Reported cumulative usage

Counter	Independent	Sharing, reported subset
Input	16,537,335	21,034,120
Cached input, included above	14,830,208	16,999,680
Output	183,706	293,098
Reasoning output, included above	24,366	21,133
Input + output	16,721,041	21,327,218

Use the last cumulative counter once per unique session. Cached input is a subset of input; reasoning is a subset of output. Nineteen sharing sessions have no counter and are unknown. Reported sums may undercount interrupted sessions, even when a final counter exists. They are not billable-cost estimates and should not be compared as complete usage totals.

Adherence and claim-quality findings

Many sharing selections used OS integers modulo 3 without rejection; exact uniformity is not guaranteed. Five plans were first written after receiving peer context, rather than independently at generation 0. Three workers stopped before producing a plan. The independent arm had four modulo-without-rejection selections and one missing plan. Nonces and fresh session IDs do not demonstrate semantic diversity.

Baseline replay, split vectors, region scheduling and clearing-cost proposals recurred. Some agents never pursued their selected improvement before a baseline succeeded; many successors repeated setup and interface discovery. A checkpoint claimed zero adjacent-XOR pairs before reading output that actually reported 818/825. Such narratives were not counted as scientific findings. Primary exact word/receipt checks are unaffected.

Methods, recovery and evidence

Final interpretation and recoverable compute shutdown

Frozen scientific contract

The same PR71 baseline task and checker mathematics were used across these two Daytona arms. Model gpt-6-astra, high reasoning effort, temperature unchanged. Initial-agent prompts requested three own hypotheses and private OS selection; actual adherence is documented on page 6. No hand-authored agent role partition. First frozen-checker valid=true, including eligible baseline, or 3600 s common group elapsed cap. Checkpoints were maintained during work; successors received literal same-group snapshots and retained their own checkout with fresh auth-only provider state. No synthesis-model call, novelty escalation, forced wait or discarded early success.

Concurrency and resource amendments

Independent: first four groups / ten workers concurrently, followed by 16 groups / 40 workers. The early four had no checker admission receipt; the remaining 16 used admission 2 on a 2 CPU / 4 GiB controller. Sharing: quota-aware parallel JIT setup, up to 108 workers concurrently, setup 2 groups and archive 4, with checker admission 4 on a 4 CPU / 8 GiB controller. Workers remained 2 CPU / 4 GiB / 8 GiB disk. Increased provider load, controller resources and queue admission differed across arms.

Failure and retry accounting

Original sharing trial 5 /4 agents /5 min failed after 71.894 s because the selected model was at capacity. Its four originals were exported and stopped, filesystem retained. The authorized fresh four-worker retry succeeded at 80.580 s submission / 90.844 s validation under lower load. Exactly 44 original successes plus the retry form 45 logical slots. The failed attempt remains a 46th attempt. Earlier SSH/control and archive lifecycle recoveries were infrastructure events; cooldown-only attempts had no research sessions and are not research failures.

Compute and recovery preservation

All 135 successful sharing workers and 50 independent workers are recoverably archived after entry/hash verification. Four failed originals are stopped and retained. Sharing controller, archive collector and independent controller were verified idle and stopped; no research remains running. No controller or archive was deleted. The sole old Mac checkout recovery archive was rehashed before shutdown and preserved with the 50 independent worker archives.

Scope and comparability

These are conditional finite-word construction checks, not theorem proofs or multiplication-runtime benchmarks. All 65 primary pairs exactly match the supplied baseline; zero requested milestone gains were verified. Historical Mac controls remain paused 18/20 and native sharing 0/45, outside this analysis. Linux platform, compiler, harness/root launch, account, cache, provider load, concurrent scheduling and checker queues confound historical and cross-arm comparisons. No causal scaling, diversity or sharing claim is warranted.

Reproducible artifacts and evidence

Per-trial data | All cell statistics | Aggregate and source hashes | Sharing final audit | Exact word / literal-join audit | Response corroboration | Owned compute terminal states

Artifacts sit beside this PDF or in its parent evidence directory. Relative evidence links work when the directory is retained. Source hashes and full precision values are in the aggregate; chart PNG/PDF exports are in final-figures/. Earlier current-report.pdf remains a dated provisional snapshot.